

Collection Risk Absorption and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Collection Risk Absorption (CRA), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CRA achieves an annualized gross (net) Sharpe ratio of 0.33 (0.26), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (17) bps/month with a t-statistic of 2.33 (1.89), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Composite debt issuance, Growth in book equity, change in net operating assets, Asset growth, Pension Funding Status, Trend Factor) is 23 bps/month with a t-statistic of 2.16.

1 Introduction

Asset pricing theory predicts that differences in expected returns across securities reflect compensation for bearing systematic risk that covaries with investors' marginal utility of consumption. While this fundamental insight from consumption-based asset pricing models provides the theoretical foundation for understanding the cross-section of returns, empirical evidence linking firm characteristics to consumption risk remains incomplete. We identify a novel channel through which firms' collection risk absorption activities create differential exposure to aggregate consumption shocks, generating predictable variation in expected returns.

The collection of accounts receivable represents a critical junction where firms absorb counterparty default risk and liquidity shocks that intensify during periods of low consumption growth and high marginal utility. Firms that extend more generous credit terms or maintain larger receivables balances effectively provide insurance to their customers against consumption volatility, bearing risks that command a premium in equilibrium. Despite the economic significance of trade credit markets—which exceed bank lending in magnitude—the asset pricing implications of collection risk absorption remain unexplored in the literature.

Collection Risk Absorption (CRA) captures firms' exposure to systematic consumption risk through their trade credit policies and receivables management practices. When firms extend credit to customers and delay collection, they bear the risk that economic downturns will impair customers' ability to pay, directly linking firm cash flows to aggregate consumption dynamics [Mian and Smith Jr \(1992\)](#). This mechanism operates through the intertemporal marginal rate of substitution, as collection losses spike precisely when investors' marginal utility is highest, making these cash flow shocks particularly painful [Campbell and Cochrane \(1999\)](#).

The state-dependent nature of collection risk creates pronounced exposure to consumption disasters and long-run risks. During economic contractions characterized

by declining consumption growth, firms with high CRA face a double burden: increased default rates on existing receivables and pressure to extend even more lenient credit terms to maintain sales [Petersen and Rajan \(1997\)](#). This procyclical variation in collection risk generates covariance with the stochastic discount factor that varies systematically with firms’ receivables management strategies. Firms absorbing more collection risk experience cash flow volatility that correlates strongly with consumption growth innovations, particularly the persistent component that drives long-run risks [Bansal and Yaron \(2004\)](#).

The consumption-based interpretation of CRA aligns with habit formation models where risk premia vary with the consumption surplus ratio. Firms with high collection risk absorption effectively sell insurance against habit shocks, as their cash flows deteriorate most severely when consumption approaches the habit level and risk aversion peaks [Campbell and Cochrane \(1999\)](#). This insurance provision manifests through extended payment terms and accumulated receivables that default en masse during consumption disasters, creating systematic risk exposure that rational investors demand compensation to bear [Barro \(2006\)](#). The resulting return predictability reflects equilibrium compensation for bearing undiversifiable consumption risk through the collection channel.

Our empirical analysis reveals that Collection Risk Absorption strongly predicts cross-sectional variation in expected returns. A value-weighted long/short trading strategy based on CRA achieves an annualized gross Sharpe ratio of 0.33, with a monthly average excess return of 21 basis points and a t-statistic of 2.40. The strategy’s alpha remains economically and statistically significant across all standard factor models, ranging from 18 to 24 basis points per month with t-statistics exceeding 1.94 throughout. Most notably, the alpha relative to the six-factor model of [Fama and French \(2018\)](#) equals 22 basis points per month with a t-statistic of 2.33.

The CRA premium exhibits remarkable consistency across different portfolio con-

struction methodologies and size groups. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the CRA strategy delivers an average return of 25 basis points per month with a t-statistic of 2.28. The strategy’s factor loadings reveal a significant negative exposure to the RMW profitability factor of -0.12 (t-statistic = -2.81), consistent with high CRA firms experiencing lower profitability due to collection losses during high marginal utility states. After accounting for transaction costs using the composite bid-ask spreads of [Chen and Velikov \(2022\)](#), the strategy maintains a net Sharpe ratio of 0.26 and significantly expands the mean-variance efficient frontier.

The economic magnitude of the CRA premium persists after controlling for closely related anomalies and the broader factor zoo. When we simultaneously control for the six factors from [Fama and French \(2018\)](#) and the six most closely related strategies—including composite debt issuance, growth in book equity, and asset growth—the CRA strategy retains an alpha of 23 basis points per month with a t-statistic of 2.16. Furthermore, incorporating CRA into combination strategies that aggregate signals from 155 documented anomalies increases terminal wealth by 36%, demonstrating that collection risk absorption captures a distinct dimension of systematic risk not spanned by existing factors.

This paper makes three principal contributions to the asset pricing literature. First, we document a novel consumption-based risk factor rooted in firms’ collection risk absorption activities, extending the empirical content of consumption-based asset pricing models. While [Jagannathan and Wang \(1996\)](#) and [Lettau and Ludvigson \(2001\)](#) identify consumption-related state variables that price the cross-section, our work demonstrates that firm-level exposure to consumption risk through the receivables channel generates significant return predictability. Unlike the investment-based factors of [Hou et al. \(2015\)](#) or the intermediary asset pricing factors of [He et al. \(2017\)](#), CRA directly links firms’ operating decisions to consumption risk exposure through

a transparent economic mechanism.

Second, we expand the boundaries of accounting-based return predictors beyond traditional measures focused on accruals and investment. Previous work by [Sloan \(1996\)](#) and [Richardson et al. \(2005\)](#) establishes that accounting distortions predict returns, while [Cooper et al. \(2008\)](#) documents asset growth effects. Our CRA measure captures a fundamentally different phenomenon—the pricing of consumption risk embedded in working capital management—that generates alphas of 22 to 24 basis points per month relative to standard factor models including these accounting-based anomalies. This finding enriches our understanding of how accounting information maps to risk exposures as emphasized by [Ball and Brown \(1968\)](#) and [Penman and Zhang \(2002\)](#).

Third, our results contribute to the broader debate about risk versus mispricing explanations for cross-sectional return predictability. The consumption-based foundation of CRA, its persistence across size groups, and its robustness to transaction costs support a rational risk-based interpretation aligned with [Fama and French \(2015\)](#) rather than behavioral explanations. The 36% improvement in combination strategy performance from incorporating CRA demonstrates that collection risk absorption captures a economically significant dimension of systematic risk previously overlooked in the factor zoo of [Harvey et al. \(2016\)](#). These findings underscore the importance of examining how firms’ operating policies create differential exposure to aggregate consumption risk, opening new avenues for understanding the cross-section of expected returns.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample

includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of changes in doubtful receivables and to align with the typical reporting frequency of these accounting items. signal construction relies on two key accounting variables. Receivables estimated doubtful (RECD) represents the allowance for doubtful accounts, which is the amount that firms estimate they will be unable to collect from their accounts receivable. This contra-asset account reflects management’s assessment of collection risk associated with credit sales. Nonop income other (NOPIO) captures other non-operating income, which includes various income items not directly related to the firm’s core operations, such as interest income, dividend income, and gains from non-operating activities. construct the Collection Risk Absorption signal by calculating the year-over-year change in receivables estimated doubtful, scaled by lagged non-operating income other: $(\text{RECD}(t) - \text{RECD}(t-1)) / \text{NOPIO}(t-1)$. This measure captures how changes in collection risk provisions relate to the firm’s non-operating income base. An increase in this ratio suggests that the firm is absorbing greater collection risk relative to its non-operating income cushion, potentially indicating deteriorating credit quality or more aggressive revenue recognition practices. We compute this signal using fiscal year-end values and require non-missing values for both RECD and NOPIO in consecutive years. To ensure meaningful ratios, we exclude observations where the denominator $\text{NOPIO}(t-1)$ equals zero or is negative, as these cases would produce economically uninterpretable values.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CRA signal. Panel A plots the time-series of the mean, median, and interquartile range for CRA. On average, the cross-

sectional mean (median) CRA is 0.13 (-0.01) over the 1971 to 2024 sample, where the starting date is determined by the availability of the input CRA data. The signal’s interquartile range spans -0.53 to 0.33. Panel B of Figure 1 plots the time-series of the coverage of the CRA signal for the CRSP universe. On average, the CRA signal is available for 3.63% of CRSP names, which on average make up 4.61% of total market capitalization.

4 Does CRA predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CRA using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CRA portfolio and sells the low CRA portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CRA strategy earns an average return of 0.21% per month with a t-statistic of 2.40. The annualized Sharpe ratio of the strategy is 0.33. The alphas range from 0.18% to 0.24% per month and have t-statistics exceeding 1.94 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is -0.12, with a t-statistic of -2.81 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 352

stocks and an average market capitalization of at least \$881 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 12 bps/month with a t-statistics of 2.56. Out of the twenty-five alphas reported in Panel A, the t-statistics for nineteen exceed two, and for one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -12-25bps/month. The lowest return, (-12 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -2.03. Out of the twenty-five

construction-methodology-factor-model pairs reported in Panel B, the CRA trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in four cases.

Table 3 provides direct tests for the role size plays in the CRA strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CRA, as well as average returns and alphas for long/short trading CRA strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CRA strategy achieves an average return of 25 bps/month with a t-statistic of 2.28. Among these large cap stocks, the alphas for the CRA strategy relative to the five most common factor models range from 25 to 30 bps/month with t-statistics between 2.14 and 2.69.

5 How does CRA perform relative to the zoo?

Figure 2 puts the performance of CRA in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CRA strategy falls in the distribution. The CRA strategy’s gross (net) Sharpe ratio of 0.33 (0.26) is greater than 70% (84%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

CRA strategy (red line).² Ignoring trading costs, a \$1 invested in the CRA strategy would have yielded \$2.38 which ranks the CRA strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CRA strategy would have yielded \$1.50 which ranks the CRA strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CRA relative to those. Panel A shows that the CRA strategy gross alphas fall between the 48 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197106 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CRA strategy has a positive net generalized alpha for five out of the five factor models. In these cases CRA ranks between the 65 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CRA add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CRA with 208 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CRA or at least to weaken the power CRA has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CRA conditioning on each of the 208 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CRA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CRA}CRA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 208 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CRA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 208 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 208 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CRA. Stocks are finally grouped into five CRA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CRA trading strategies conditioned on each of the 208 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CRA and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CRA signal in these Fama-MacBeth regressions exceed 0.11, with the minimum t-statistic occurring when controlling for Composite debt issuance. Controlling for all six closely related anomalies, the t-statistic on CRA is 0.74.

Similarly, Table 5 reports results from spanning tests that regress returns to the CRA strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CRA strategy earns alphas that range from 19-28bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.98, which is achieved when controlling for Composite debt issuance. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CRA trading strategy achieves an alpha of 23bps/month with a t-statistic of 2.16.

7 Does CRA add relative to the whole zoo?

Finally, we can ask how much adding CRA to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the CRA signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CRA is available.

average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$1083.64, while \$1 investment in the combination strategy that includes CRA grows to \$1474.35.

8 Conclusion

This study provides compelling evidence that Collection Risk Absorption (CRA) represents a meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CRA exhibits both statistical significance and economic relevance in asset pricing contexts. The value-weighted long/short strategy based on CRA delivers a respectable annualized net Sharpe ratio of 0.26, which, while modest, remains economically meaningful after accounting for transaction costs. More notably, the signal generates significant abnormal returns even after controlling for established risk factors, with monthly alphas of 17 basis points (net) relative to the Fama-French five-factor model plus momentum, and maintains its significance when further controlling for six closely related strategies from the factor zoo.

The persistence of CRA’s predictive power, even after accounting for related signals such as composite debt issuance, asset growth, and pension funding status, suggests that it captures a distinct dimension of firm-level risk absorption that is not fully subsumed by existing factors. This finding has important practical implications for both academic researchers and investment practitioners. For academics, CRA offers a new lens through which to examine how firms’ collection practices and credit risk management strategies influence their market valuations. For practitioners, the signal presents a potentially exploitable investment strategy that appears robust to standard risk adjustments and related anomalies.

However, several limitations of this study warrant consideration. First, while

the net Sharpe ratio of 0.26 indicates profitability after transaction costs, the strategy’s performance may be sensitive to implementation details, market conditions, and capacity constraints not fully captured in our analysis. Second, the relatively modest t-statistics (1.89 for net returns) suggest that while statistically significant, the signal’s strength may vary across different sample periods or market regimes. Third, our analysis focuses primarily on U.S. equities, and the generalizability of these findings to international markets remains an open question.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the CRA anomaly could provide deeper insights into why markets appear to misprice firms based on their collection risk absorption practices. Second, examining the signal’s performance across different market conditions, particularly during credit crises or economic downturns, would enhance our understanding of its risk-return profile. Third, testing CRA in international markets could reveal whether this phenomenon is unique to U.S. markets or represents a more universal pricing anomaly. Finally, exploring potential interactions between CRA and other corporate financial policies, such as capital structure decisions or working capital management, may uncover additional sources of predictive power and help refine the signal’s construction. Despite its limitations, this study establishes Collection Risk Absorption as a valuable addition to the growing literature on cross-sectional return predictability and highlights the continued importance of examining how firms’ operational and financial risk management decisions influence their market valuations.

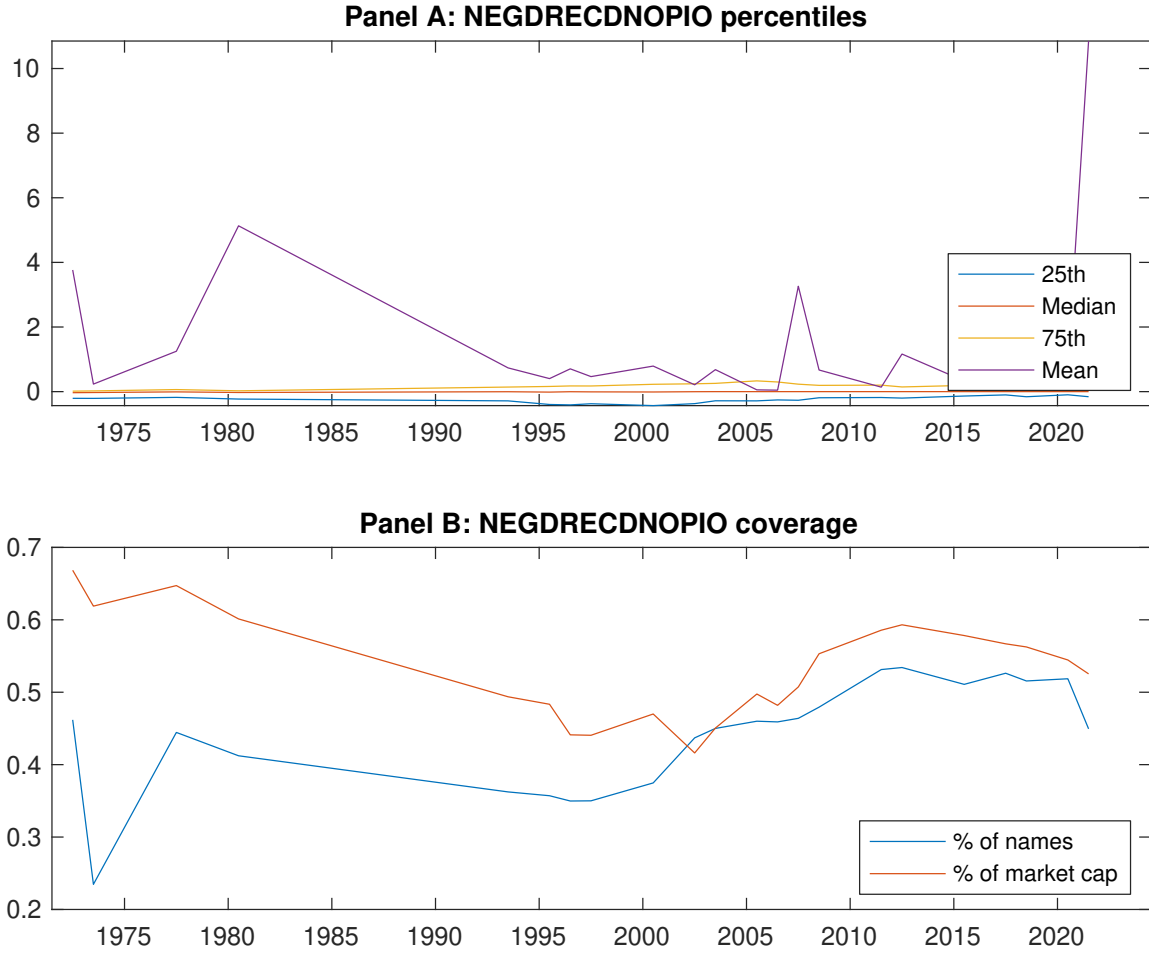


Figure 1: Times series of CRA percentiles and coverage.
This figure plots descriptive statistics for CRA. Panel A shows cross-sectional percentiles of CRA over the sample. Panel B plots the monthly coverage of CRA relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CRA. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197106 to 202406.

Panel A: Excess returns and alphas on CRA-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.55 [2.62]	0.62 [3.15]	0.55 [2.84]	0.74 [4.05]	0.76 [3.68]	0.21 [2.40]
α_{CAPM}	-0.13 [-2.03]	-0.03 [-0.52]	-0.07 [-0.97]	0.15 [2.38]	0.09 [1.32]	0.23 [2.54]
α_{FF3}	-0.08 [-1.29]	-0.02 [-0.29]	-0.04 [-0.56]	0.13 [2.20]	0.12 [1.75]	0.20 [2.24]
α_{FF4}	-0.08 [-1.30]	0.03 [0.52]	0.00 [0.00]	0.15 [2.48]	0.09 [1.36]	0.18 [1.94]
α_{FF5}	-0.10 [-1.57]	-0.02 [-0.32]	0.04 [0.62]	0.07 [1.21]	0.14 [1.95]	0.24 [2.59]
α_{FF6}	-0.10 [-1.57]	0.02 [0.37]	0.07 [0.96]	0.10 [1.63]	0.11 [1.62]	0.22 [2.33]
Panel B: Fama and French (2018) 6-factor model loadings for CRA-sorted portfolios						
β_{MKT}	1.06 [69.67]	1.04 [75.90]	0.97 [56.52]	1.00 [69.59]	1.03 [62.71]	-0.03 [-1.19]
β_{SMB}	0.13 [5.60]	-0.03 [-1.30]	-0.07 [-2.75]	-0.14 [-6.47]	0.18 [7.30]	0.05 [1.62]
β_{HML}	-0.16 [-5.43]	-0.07 [-2.86]	-0.00 [-0.04]	-0.07 [-2.64]	-0.07 [-2.35]	0.08 [2.03]
β_{RMW}	0.07 [2.38]	-0.00 [-0.17]	-0.12 [-3.54]	0.01 [0.30]	-0.05 [-1.49]	-0.12 [-2.81]
β_{CMA}	-0.01 [-0.33]	0.05 [1.35]	-0.19 [-3.72]	0.28 [6.78]	-0.01 [-0.12]	0.01 [0.14]
β_{UMD}	0.00 [0.09]	-0.06 [-4.24]	-0.04 [-2.16]	-0.04 [-2.63]	0.03 [1.92]	0.03 [1.40]
Panel C: Average number of firms (n) and market capitalization (me)						
n	433	352	415	354	429	
me (\$10 ⁶)	1059	1438	1497	1467	881	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CRA strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197106 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.21 [2.40]	0.23 [2.54]	0.20 [2.24]	0.18 [1.94]	0.24 [2.59]	0.22 [2.33]
Quintile	NYSE	EW	0.12 [2.56]	0.14 [2.91]	0.14 [2.95]	0.13 [2.64]	0.15 [3.08]	0.14 [2.82]
Quintile	Name	VW	0.22 [2.38]	0.23 [2.39]	0.20 [2.08]	0.19 [1.97]	0.24 [2.44]	0.23 [2.33]
Quintile	Cap	VW	0.18 [2.06]	0.21 [2.37]	0.16 [1.90]	0.14 [1.62]	0.16 [1.84]	0.15 [1.62]
Decile	NYSE	VW	0.30 [2.51]	0.29 [2.34]	0.28 [2.28]	0.29 [2.30]	0.36 [2.91]	0.36 [2.87]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.17 [1.87]	0.18 [2.03]	0.16 [1.74]	0.15 [1.60]	0.18 [1.98]	0.17 [1.89]
Quintile	NYSE	EW	-0.12 [-2.03]					
Quintile	Name	VW	0.17 [1.84]	0.18 [1.85]	0.15 [1.55]	0.15 [1.53]	0.17 [1.78]	0.17 [1.78]
Quintile	Cap	VW	0.14 [1.57]	0.17 [1.90]	0.13 [1.45]	0.11 [1.30]	0.12 [1.37]	0.12 [1.31]
Decile	NYSE	VW	0.25 [2.02]	0.24 [1.93]	0.23 [1.85]	0.24 [1.91]	0.29 [2.33]	0.30 [2.37]

Table 3: Conditional sort on size and CRA

This table presents results for conditional double sorts on size and CRA. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CRA. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CRA and short stocks with low CRA. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197106 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CRA Quintiles					CRA Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.71 [2.53]	0.76 [2.89]	0.66 [2.43]	0.97 [3.64]	0.86 [3.14]	0.15 [1.73]	0.17 [1.99]	0.17 [1.95]	0.14 [1.63]	0.14 [1.59]	0.12 [1.37]
	(2)	0.76 [2.90]	0.82 [3.23]	0.77 [3.05]	0.80 [3.18]	0.83 [3.24]	0.07 [0.74]	0.08 [0.94]	0.09 [1.06]	0.13 [1.44]	0.14 [1.49]	0.16 [1.77]
	(3)	0.77 [3.07]	0.81 [3.44]	0.66 [2.73]	0.87 [3.72]	0.77 [3.15]	0.00 [0.04]	0.02 [0.16]	0.04 [0.36]	0.04 [0.41]	0.11 [1.05]	0.11 [1.06]
	(4)	0.68 [3.02]	0.79 [3.38]	0.74 [3.31]	0.75 [3.39]	0.74 [3.03]	0.06 [0.60]	0.00 [0.02]	0.01 [0.14]	-0.01 [-0.07]	0.10 [1.00]	0.07 [0.76]
	(5)	0.46 [2.25]	0.58 [2.97]	0.51 [2.57]	0.76 [4.08]	0.72 [3.72]	0.25 [2.28]	0.30 [2.69]	0.26 [2.31]	0.25 [2.23]	0.25 [2.18]	0.25 [2.14]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CRA Quintiles					CRA Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	213	214	214	213	212	18	20	17	20	18	
	(2)	64	64	64	64	64	36	37	36	37	36	
	(3)	45	46	46	46	46	64	64	63	64	65	
	(4)	39	38	38	39	39	142	147	143	145	140	
(5)	35	35	35	35	35	873	1115	1187	1100	754		

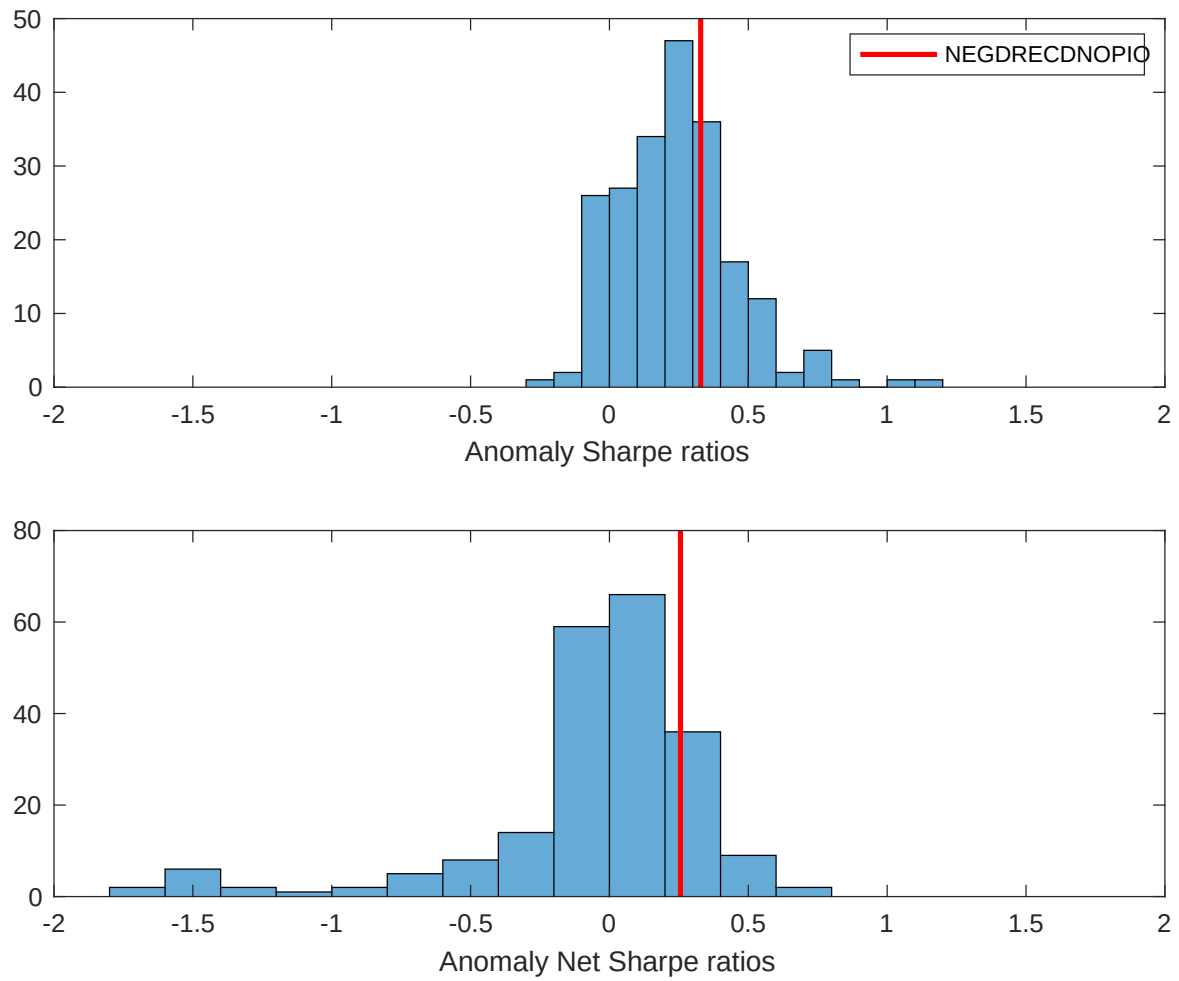


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CRA with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

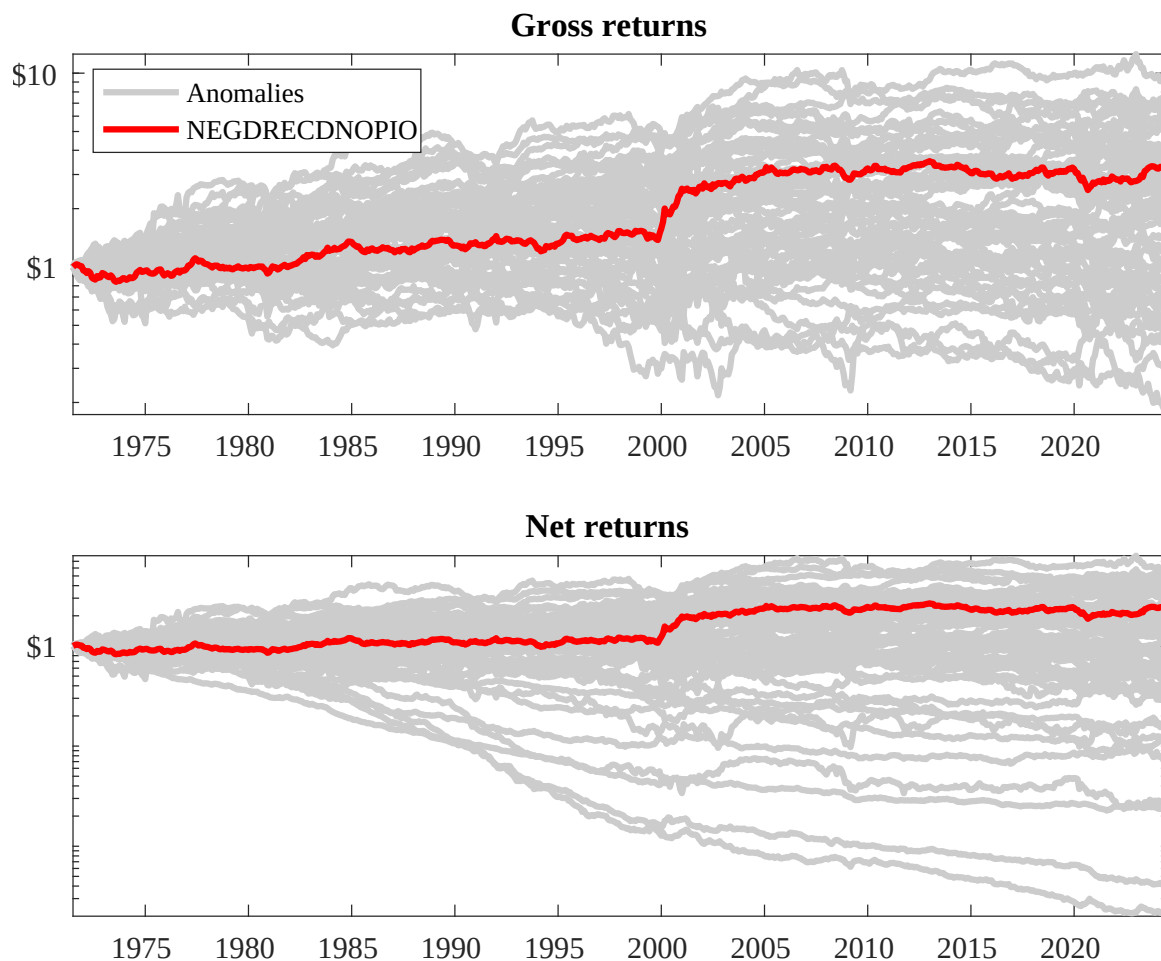
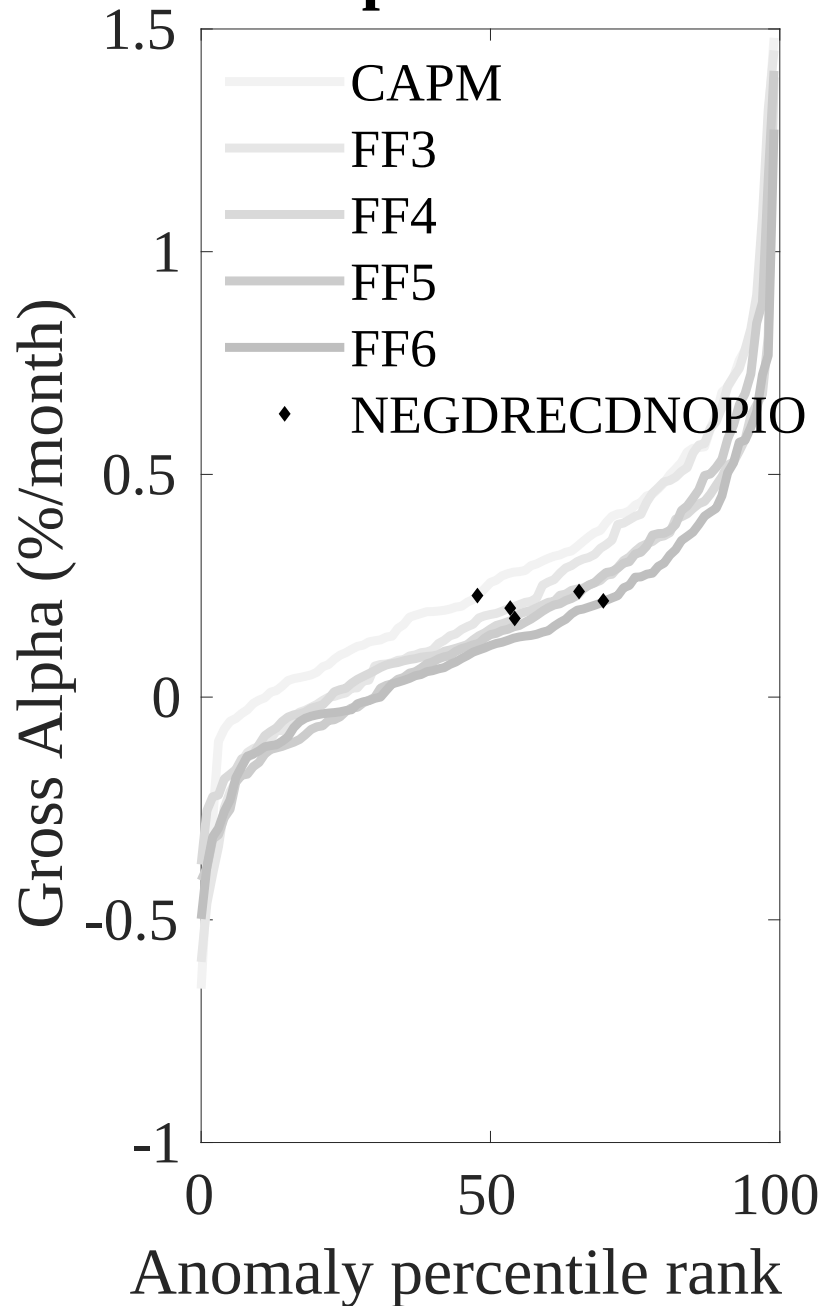


Figure 3: Dollar invested.

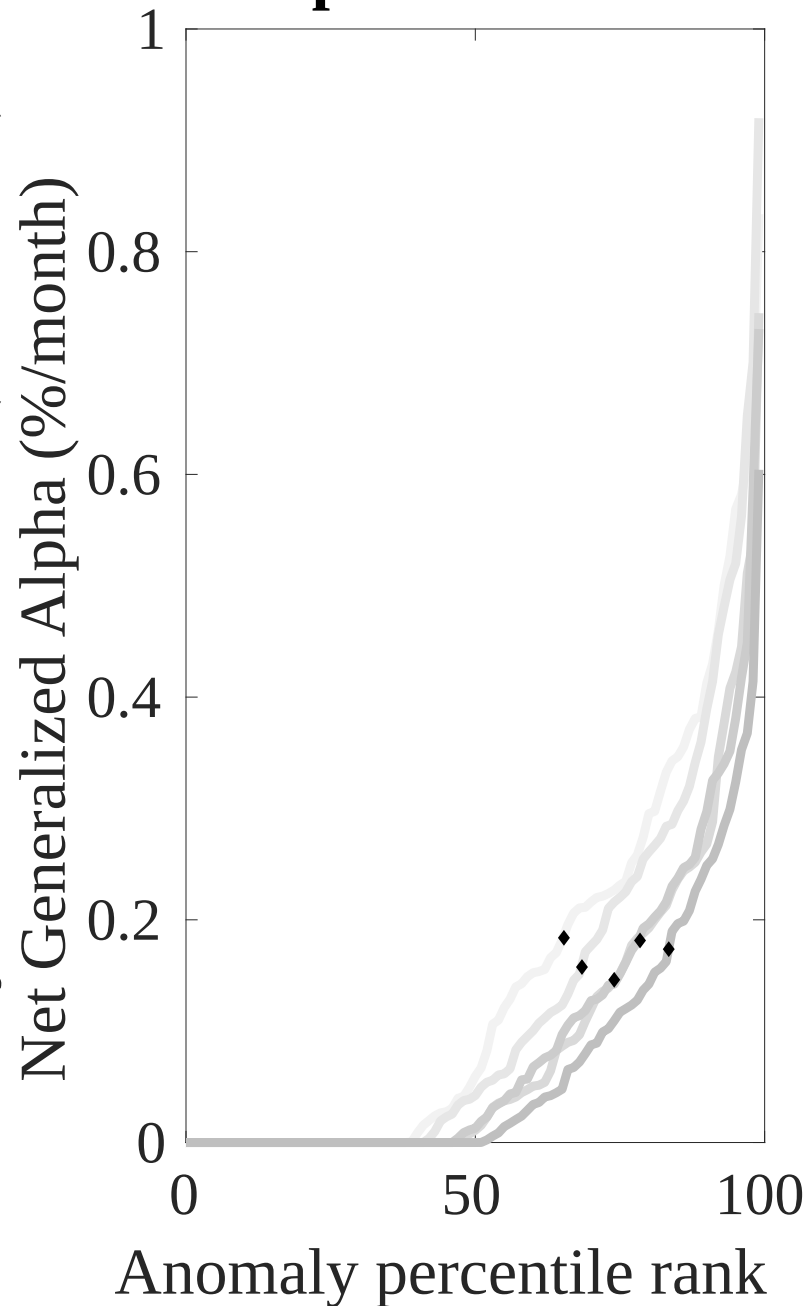
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CRA trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



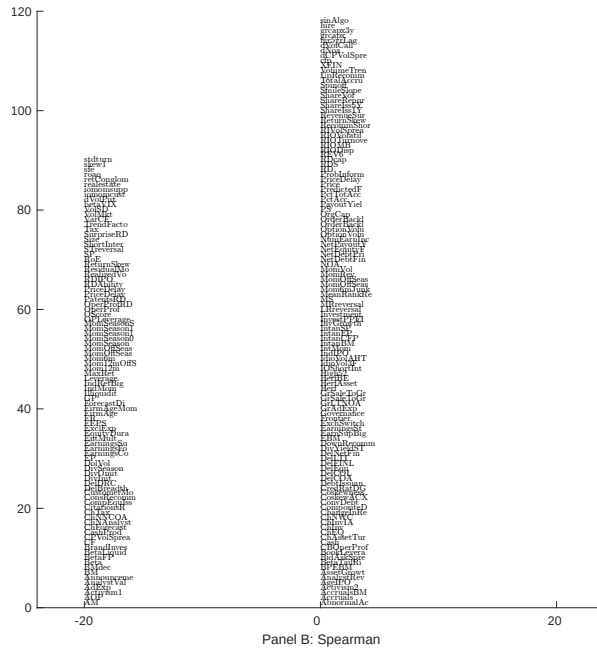
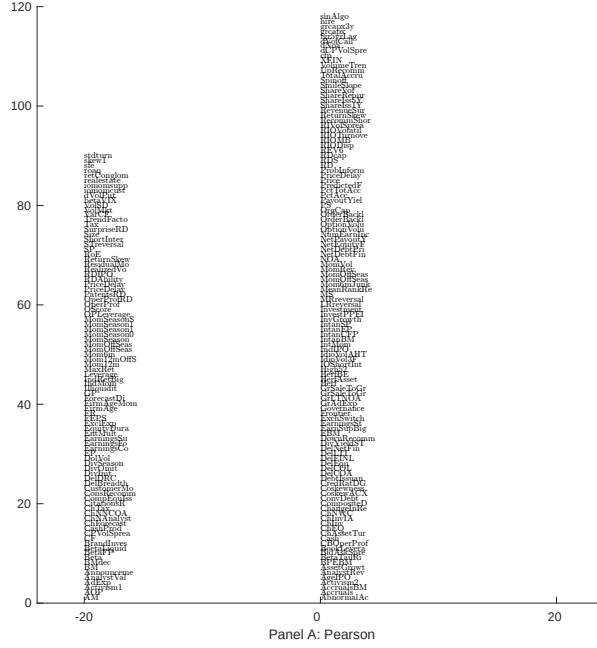


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 208 filtered anomaly signals with CRA. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

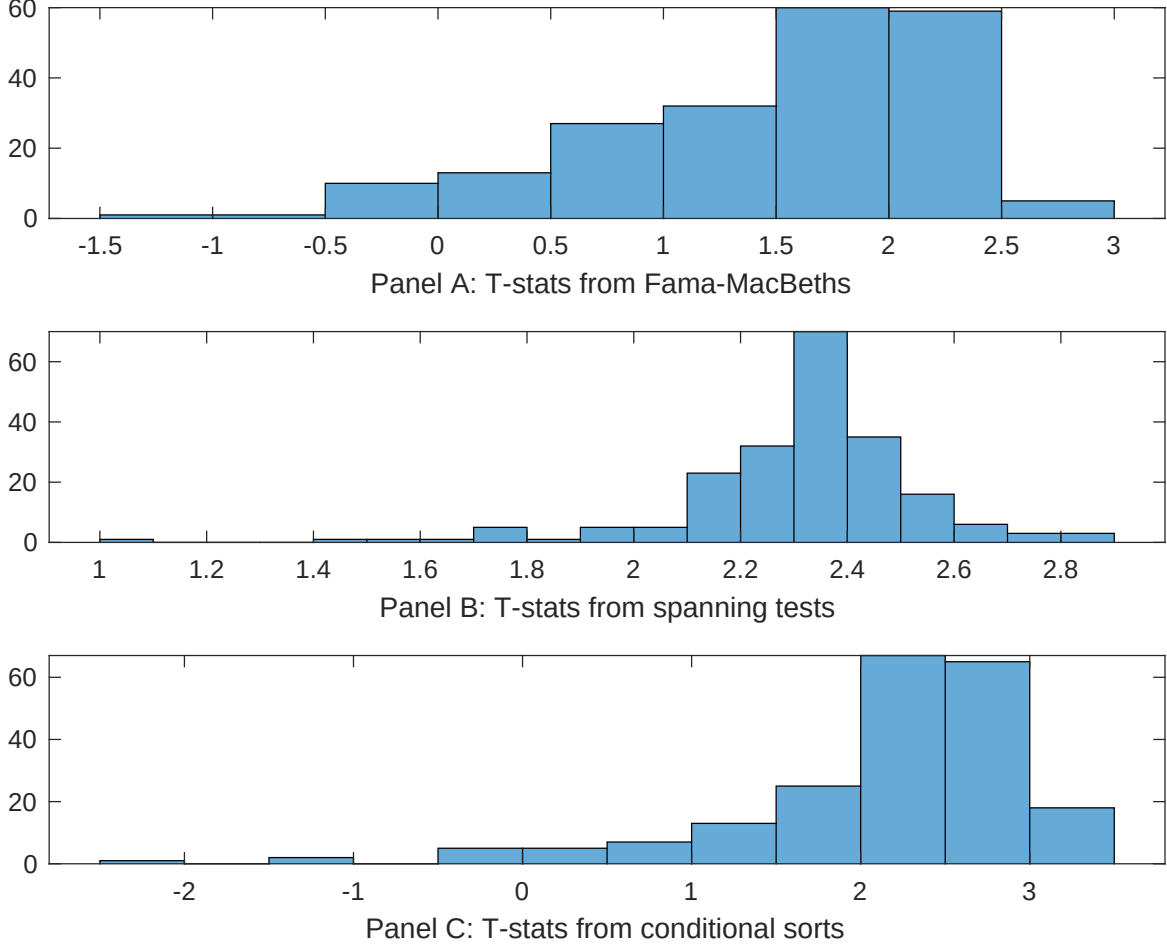


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CRA conditioning on each of the 208 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CRA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CRA}CRA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 208 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CRA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 208 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 208 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CRA. Stocks are finally grouped into five CRA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CRA trading strategies conditioned on each of the 208 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CRA. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CRA}CRA_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Composite debt issuance, Growth in book equity, change in net operating assets, Asset growth, Pension Funding Status, Trend Factor. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197106 to 202406.

Intercept	0.13 [5.61]	0.18 [6.70]	0.13 [5.42]	0.14 [5.51]	0.12 [5.05]	-0.99 [-9.48]	-0.98 [-7.17]
CRA	0.67 [0.11]	0.90 [1.84]	0.69 [1.41]	0.74 [1.49]	0.12 [1.50]	0.39 [0.71]	0.59 [0.74]
Anomaly 1	0.11 [5.76]						0.32 [1.40]
Anomaly 2		0.51 [4.61]					0.29 [0.28]
Anomaly 3			0.15 [9.65]				0.47 [1.50]
Anomaly 4				0.10 [8.16]			-0.22 [-0.92]
Anomaly 5					0.60 [1.06]		0.51 [0.90]
Anomaly 6						0.47 [8.01]	0.49 [5.99]
# months	636	636	636	636	516	631	516
$\bar{R}^2(\%)$	0	0	0	0	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CRA trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CRA} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Composite debt issuance, Growth in book equity, change in net operating assets, Asset growth, Pension Funding Status, Trend Factor. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197106 to 202406.

Intercept	0.22 [2.38]	0.21 [2.28]	0.21 [2.20]	0.22 [2.33]	0.28 [2.59]	0.19 [1.98]	0.23 [2.16]
Anomaly 1	10.47 [2.40]						14.54 [2.92]
Anomaly 2		14.40 [2.86]					22.14 [3.20]
Anomaly 3			8.55 [1.58]				3.71 [0.57]
Anomaly 4				4.39 [0.80]			-13.51 [-1.82]
Anomaly 5					-11.12 [-2.87]		-7.44 [-1.91]
Anomaly 6						3.34 [1.58]	5.55 [2.39]
mkt	-1.88 [-0.86]	-2.05 [-0.95]	-2.46 [-1.13]	-2.50 [-1.15]	-2.84 [-1.11]	-2.81 [-1.29]	-0.96 [-0.37]
smb	3.99 [1.20]	5.26 [1.62]	5.80 [1.78]	5.37 [1.63]	4.24 [1.07]	5.60 [1.70]	2.22 [0.56]
hml	7.38 [1.76]	7.12 [1.70]	8.11 [1.94]	8.66 [2.08]	8.80 [1.88]	8.97 [2.15]	5.23 [1.11]
rmw	-12.82 [-3.02]	-11.27 [-2.67]	-11.72 [-2.76]	-11.85 [-2.79]	-9.83 [-2.03]	-10.58 [-2.45]	-8.63 [-1.77]
cma	-2.21 [-0.35]	-13.47 [-1.69]	-5.84 [-0.78]	-4.86 [-0.53]	-6.66 [-0.95]	0.26 [0.04]	-18.64 [-1.73]
umd	2.66 [1.23]	2.73 [1.27]	2.81 [1.30]	3.18 [1.47]	5.20 [2.11]	2.45 [1.11]	2.36 [0.94]
# months	636	636	636	636	517	632	517
$\bar{R}^2(\%)$	4	4	3	3	3	3	6

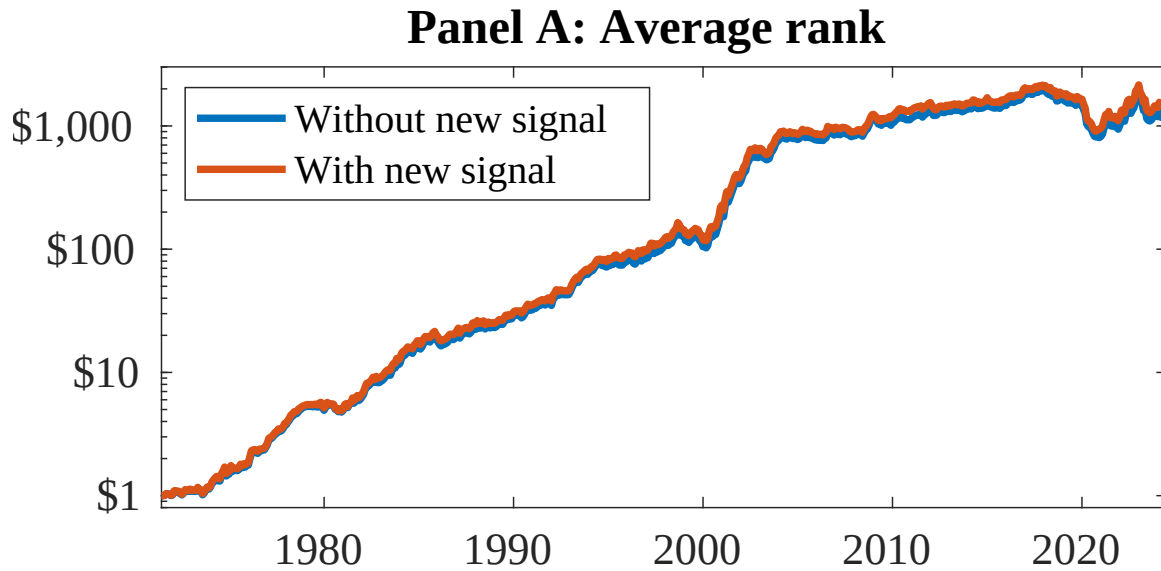


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as CRA. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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